AD7190

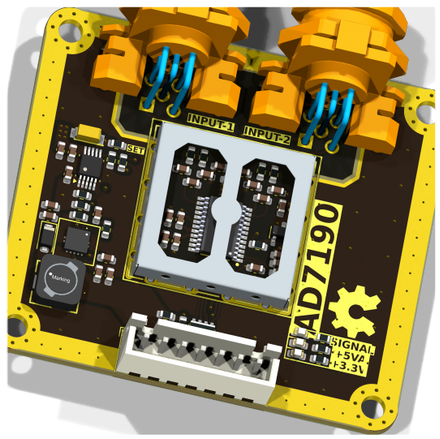
Variant:

2026-10-15

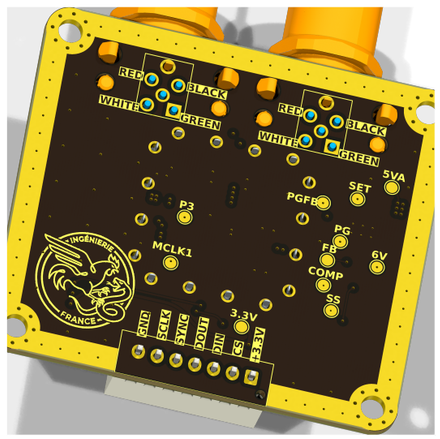
Rev + (Unreleased)

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4	AD7190_Weight_Sensor	14	.....	24	.....	34	.....
5	LT3042_6V_to_5VA	15	.....	25	.....	35	.....
6	TPS61086_3.3V_to_6V	16	.....	26	.....	36	.....
7	Holes, Fiducials	17	.....	27	.....	37	.....
8	Connectors	18	.....	28	.....	38	.....
9	Power - Sequencing	19	.....	29	.....	39	.....
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TOP VIEW



BOTTOM VIEW



NOTES

OPEN_WEIGHT - AD7190 24-bit load-cell front-end, part of OPEN THRUST LAB.
Single +3.3 V host supply, SPI interface, 2 x 4-wire bridge inputs.

Not fitted components are marked as X

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 15-Oct-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
+3.3V host -> TPS61086
6 V -> LT3042 +5VA (AVDD + excitation)

DEBUG NOTE:
TP on every rail.
Wait >= 600 ms (PG) before ADC init.

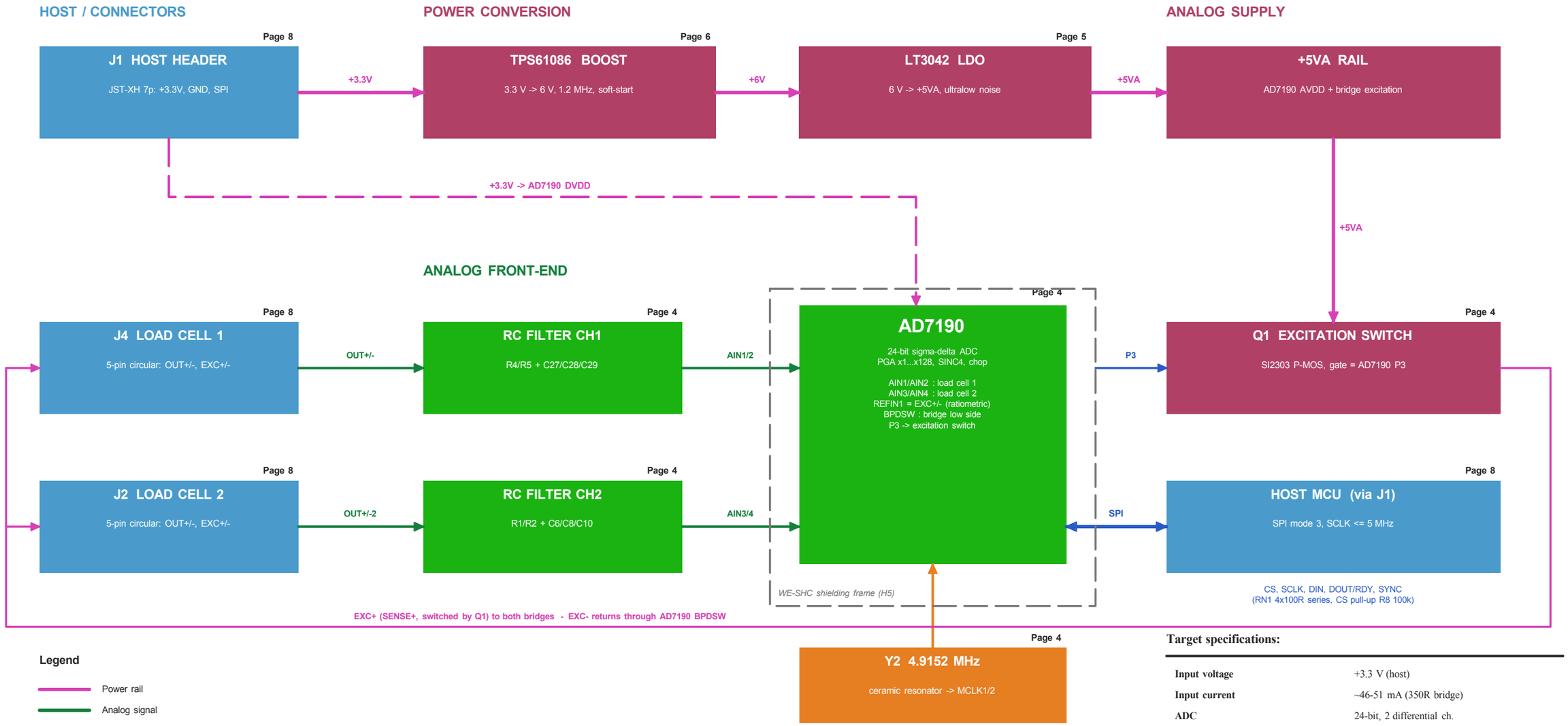
DESIGN NOTE:
SPI mode 3, SCLK <= 5 MHz, RDY on DOUT.

DESIGN NOTE:
REFIN1 = bridge excitation: ratiometric.

LAYOUT NOTE:
AFE under the WE-SHC shield H5, boost SW node away.

Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: 349b121
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
Sheet Title: OPEN_WEIGHT - AD7190	File Name: OPEN_WEIGHT.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 10

[2] Block Diagram



Legend

- Power rail
- Analog signal
- Digital / SPI
- Clock

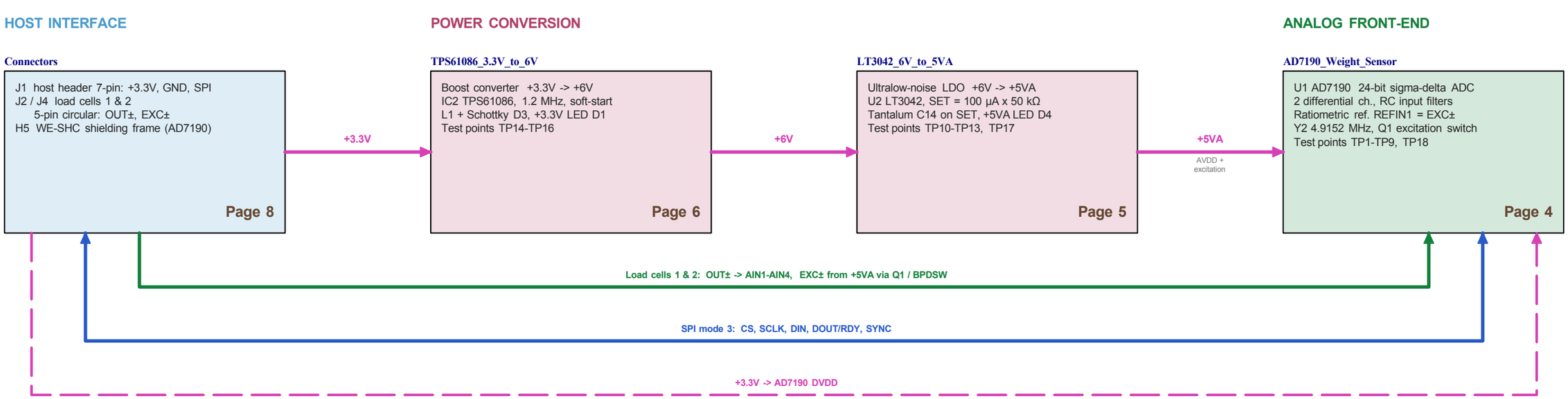
Target specifications:

Input voltage	+3.3 V (host)
Input current	~46-51 mA (350R bridge)
ADC	24-bit, 2 differential ch.
Reference	ratiometric, EXC = +5VA
Board size	57.0 x 50.75 mm, 4 layers

Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: 349b121
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: A3	Sheet: 2 of 10

[3] Project Architecture

Sheet hierarchy and signal flow. Click a block or its page number to open the sheet.



MECHANICAL

Holes, Fiducials

H1-H4 mounting holes
FID1-FID6 fiducials
No electrical function

Page 7

LEGEND

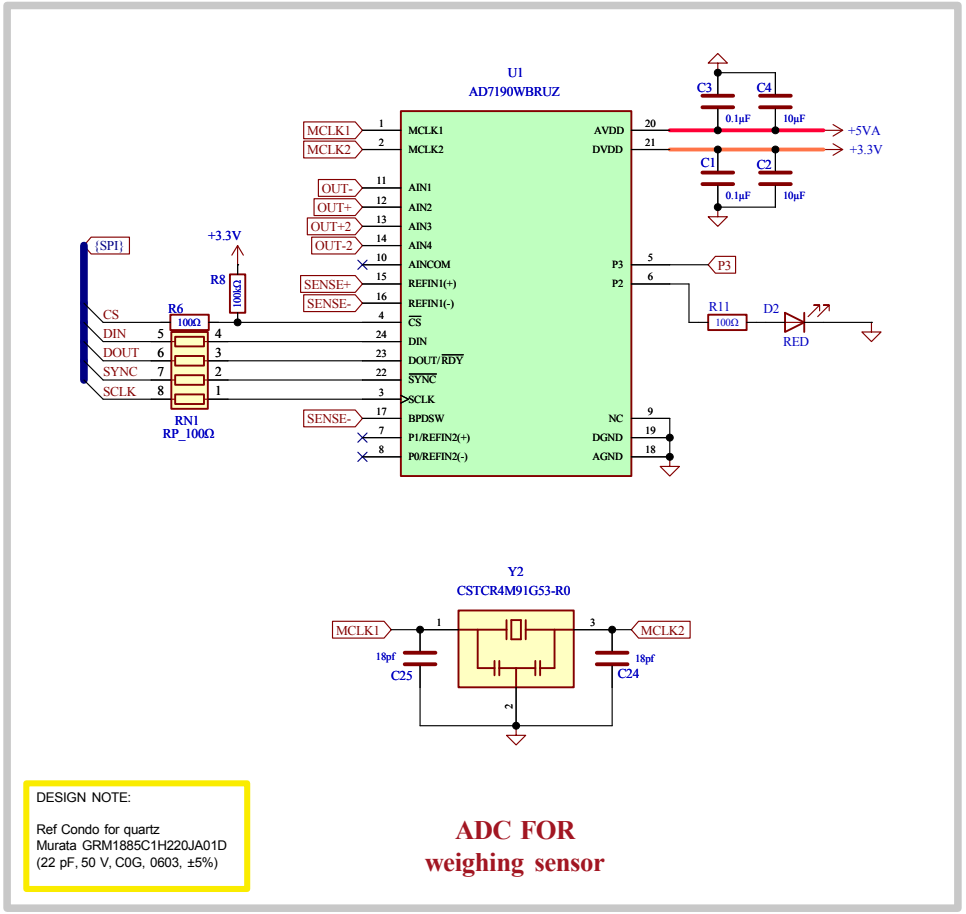
- Power rail
- Secondary supply
- Analog signal
- Digital / SPI

DOCUMENT INDEX

[1] Title Page
[2] Block Diagram
[3] Project Architecture
[4] AD7190 Weight Sensor
[5] LT3042 LDO 6V -> 5VA
[6] TPS61086 Boost 3V3 -> 6V
[7] Holes, Fiducials
[8] Connectors
[9] Power Sequencing
[10] Revision History

Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: 349b121
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 10

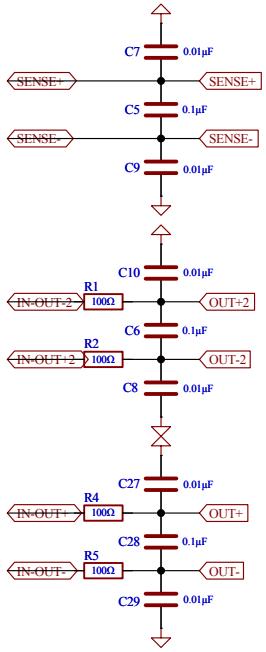
[4] AD7190 - Weight Sensor



DESIGN NOTE:

Ref Condo for quartz
Murata GRM1885C1H220JA01D
(22 pF, 50 V, C0G, 0603, ±5%)

ADC FOR weighing sensor



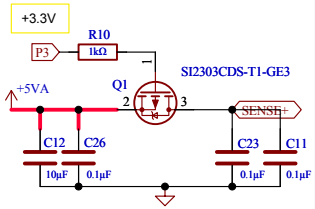
Calcule de filtre : fc = 1 / (2π × R × C) = X Hz

DESIGN NOTE:

Ref Condo for Filtre
Murata GRM1885C1H103JA01D (10 nF, 50 V, C0G, 0603, ±5%)
KEMET C0603R104K5RACTU (100 nF, 50 V, X7R, 0603, ±10%)

Ref Resistor for Filtre
TNPW0603100RBEEA (100 Ω, 0.1 W, thin film, 0603, ±0.1%, 25 ppm/°C)

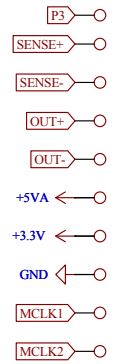
FILTRE



DESIGN NOTE:

IC1 is a P-MOS high-side switch used to enable/disable the load cell excitation.
must disabled during calibration

DRIVE GATE



TEST POINT

About

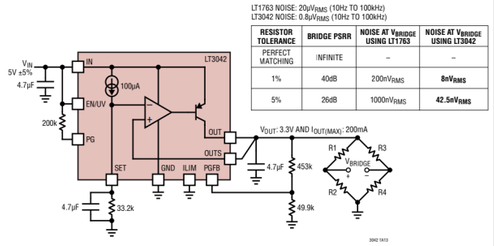
The AD7190 is a low noise, complete analog front end for high precision measurement applications. It contains a low noise, 24-bit sigma-delta (∑-Δ) analog-to-digital converter (ADC).
The on-chip low noise gain stage means that signals of small amplitude can be interfaced directly to the ADC.
The device can be configured to have two differential inputs or four pseudo differential inputs. The on-chip channel sequencer allows several channels to be enabled, and the AD7190 sequentially converts on each enabled channel. This simplifies communication with the part. The on-chip 4.92 MHz clock can be used as the clock source to the ADC or, alternatively, an external clock or crystal can be used. The output data rate from the part can be varied from 4.7 Hz to 4.8 kHz.
The device has two digital filter options. The choice of filter affects the rms noise/noise-free resolution at the programmed output data rate, the settling time, and the 50 Hz/60 Hz rejection.

Comments: AD7190 / CSTCR4M91G53-R0 / AD7190_TEST_BOARD CN	Company: OPEN THRUST LAB		Variant:	Git Hash: 349b121
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
Sheet Title: AD7190 - Weight Sensor	File Name: AD7190 - Weight Sensor.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Project Architecture/AD7190_Weight_Sensor/		Reviewer:	Size: A4	Sheet: 4 of 10

[5] LT3042 - LDO
6V to 5V Analog

TYPICAL APPLICATIONS

Low Noise Wheatstone Bridge Power Supply



LT1763 NOISE: $20\mu V_{RMS}$ (10Hz TO 100kHz)
LT3042 NOISE: $0.8\mu V_{RMS}$ (10Hz TO 100kHz)

RESISTOR TOLERANCE	BRIDGE PSRR	NOISE AT V_{BRIDGE} USING LT1763	NOISE AT V_{BRIDGE} USING LT3042
PERFECT MATCHING	INFINITE	–	–
1%	40dB	$200nV_{RMS}$	**$8nV_{RMS}$**
5%	26dB	$1000nV_{RMS}$	**$42.5nV_{RMS}$**

IN (pins 1-2): LT3042 power input (6V), to be decoupled with CIN.

EN/UV (pin 3): enables/disables the regulator and allows an UVLO threshold (often tied to IN for "always ON").

PG (pin 4): open-collector Power-Good output (requires a pull-up resistor).

ILIM (pin 5): current limit programming/monitoring (often tied to GND if unused).

PGFB (pin 6): input that sets the Power-Good threshold via a resistor divider from VOUT.

SET (pin 7): sets the output voltage via RSET (and CSET reduces noise / provides soft-start).

GND (pin 8): circuit ground reference.

OUTS (pin 9): Kelvin output sense (to be connected to the + of COUT / load point).

OUT (pin 10): regulated output (5V), to be decoupled with COUT.

EPAD (exposed pad): ground + thermal dissipation (to be soldered to the GND plane).

DESIGN NOTE:

Ref Condo for SET PIN (important pour les vibrations P16)
KYOCERA AVX TAJA225K016RNJ (2.2 µF, 16 VDC, condensateur tantale solide SMD moulé, 1206 / 3216 (A case), ±10%, ESR 6.5 Ω).

VOUT ≈ ISET × RSET
ISET ≈ 100 µA

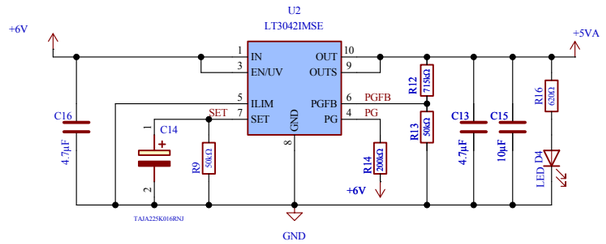
Donc :
RSET ≈ VOUT / ISET
RSET ≈ 5.0 V / 100 µA = 50 kΩ
(=> valeur standard recommandée : 49.9 kΩ)

VOUT(PG_THRESHOLD) ≈ 0.3 V × (1 + RPG2/RPG1)
PGFB_THRESHOLD ≈ 0.3 V

Donc :
RPG2 ≈ RPG1 × (VOUT(PG)/0.3 V − 1)

Exemple (PG OK à ~4.8 V) :
Choix RPG1 = 49.9 kΩ
RPG2 ≈ 49.9 kΩ × (4.8/0.3 − 1) ≈ 748.5 kΩ
(=> valeur standard recommandée : 750 kΩ)

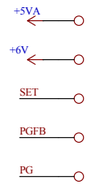
Design guide line



Convertisseur 5VA

About

The LT®3042 is a high performance low dropout linear regulator featuring LTC's ultralow noise and ultrahigh PSRR architecture for powering noise sensitive RF applications. Designed as a precision current reference followed by a high performance voltage buffer, the LT3042 can be easily paralleled to further reduce noise, increase output current and spread heat on the PCB. The device supplies 200mA at a typical 350mV dropout voltage. Operating quiescent current is nominally 2mA and drops to <<1µA in shutdown. The LT3042's wide output voltage range (0V to 15V) while maintaining unitygain operation provides virtually constant output noise, PSRR, bandwidth and load regulation, independent of the programmed output voltage. Additionally, the regulator features programmable current limit, fast start-up capability and programmable power good to indicate output voltage regulation. The LT3042 is stable with a minimum 4.7µF ceramic output capacitor. Built-in protection includes reverse battery protection, reverse current protection, internal current limit with foldback and thermal limit with hysteresis. The LT3042 is available in thermally enhanced 10-Lead MSOP and 3mm × 3mm DFN package



TEST POINT

Comments: LT3042 Analog Devices	Company: OPEN THRUST LAB		Variant:	Git Hash: 349b121
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
Sheet Title: LT3042 - LDO 6V to 5V Analog	File Name: LT3042 - LDO 6V to 5VA.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Project Architecture/LT3042_6V_to_5VA/		Reviewer:	Size: A3	Sheet: 5 of 10

[6] TPS61086 - Boost 3.3V to 6V

COMP (pin 1): error amplifier compensation pin; connect the Rcomp + Ccomp network to AGND to stabilize the loop (do not route it near SW).

FB (pin 2): feedback input; connect it to the R1/R2 resistor divider from VOUT (after the diode) to set the output voltage (e.g. 6 V).

EN (pin 3): enable / shutdown; high level = active, low level = off. Often tied to IN for "always ON", or driven by a GPIO.

AGND (pin 4): analog ground (reference for sensitive signals: FB/COMP/SS/MODE). Keep it "clean" and connect it to the GND plane in a controlled way.

PGND (pin 5): power ground (return path for switching currents). Connect it to the GND plane with a short/wide copper area close to the power capacitors.

SW (pins 6-7): switch node; connect it to L and the Schottky diode. This is the noisiest node → keep the trace short, minimize the loop area, and keep it away from FB/COMP.

IN (pin 8): power input (2.3-6 V); decouple it as close as possible with Cby ~1 µF + Cin (e.g. 10 µF) to GND.

MODE (pin 9): mode selection; MODE = High → Forced PWM (more stable frequency, better for noise), MODE = Low → PFM (better efficiency at light load).

SS (pin 10): soft-start; add a capacitor (Css) to limit inrush current at startup. Left open = no soft-start.

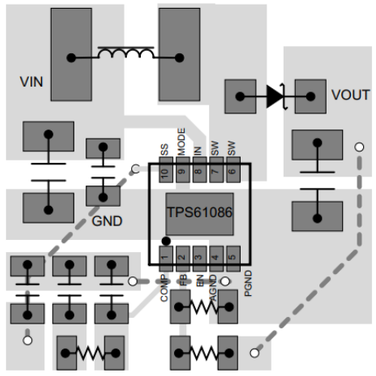
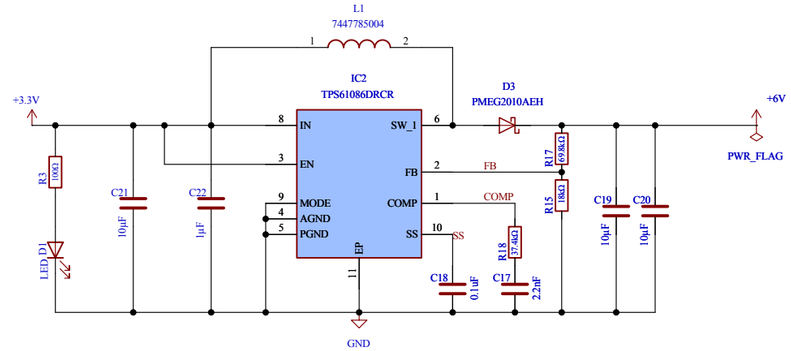


Figure 26. TPS61086 Layout Example

Pin usage and layout example



Boost 3.3 V -> 6.04 V

About

The TPS61086 device is a high-frequency, high-efficiency DC-to-DC converter with an integrated 2.0-A, 0.13-Ω power switch capable of providing an output voltage up to 18.5 V. The implemented boost converter is based on a fixed frequency of 1.2-MHz, pulse-width-modulation (PWM) controller that allows the use of small external inductors and capacitors and provides fast transient response.

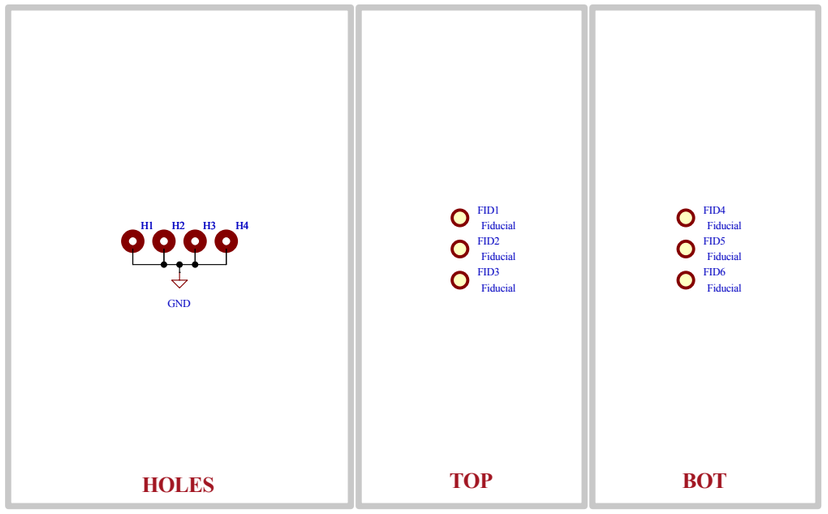
At light-load, the device can operate in Power Save Mode with pulse-frequency-modulation (PFM) to improve the efficiency while keeping a low-output voltage ripple. For very noise-sensitive applications, the device can be forced to PWM Mode operation over the entire load range by pulling the MODE pin high. The external compensation allows optimizing the application for specific conditions. A capacitor connected to the soft-start pin minimizes inrush current at start-up.



Test points

Comments: **References:** TI TPS61086 datasheet; L1 7447785004, D3 PMEG2010AEH	Company: **OPEN THRUST LAB**		Variant:	Git Hash: 349b121
	Board Name: **AD7190**		Project Name: **OPEN_WEIGHT**	
Sheet Title: TPS61086 - Boost 3.3V to 6V	File Name: TPS61086 - Boost 3V3 to 6V.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Project Architecture/TPS61086_3.3V_to_6V/		Reviewer:	Size: **A3**	Sheet: **6** of **10**

[7] Holes, Fiducials



Comments: **References:** 4 x M3 (3.2 mm) plated holes to GND FID1-FID6 not fitted on this revision	Company: **OPEN THRUST LAB**		Variant:	Git Hash: 349b121
	Board Name: **AD7190**		Project Name: **OPEN_WEIGHT**	
Sheet Title: Holes, Fiducials	File Name: Holes - Fiducials.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Holes, Fiducials/		Reviewer:	Size: **A3**	Sheet: **7** of **10**

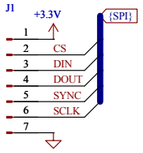
[8] Connectors

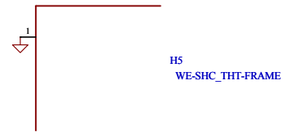
DESIGN NOTE:
J1 host header (JST-XH, 7 pins):
1 +3.3V 2 CS 3 DIN 4 DOUT/RDY
5 SYNC 6 SCLK 7 GND

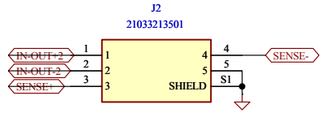
DESIGN NOTE:
J2/J4 load cell (5 pins):
1 OUT, 2 OUT (signal pair), 3 EXC+ (SENSE+),
4 EXC- (SENSE-/BPDSW), 5 GND/shield.
Both channels read V(pin2) - V(pin1):
fix the sign in firmware if needed.

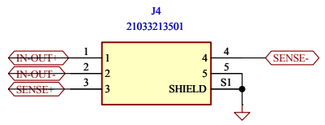
LAYOUT NOTE:
H5 shielding frame covers the AD7190
front-end; connector shells tied to GND.

Pinout and notes









Host and load-cell connectors





Power flags: +3.3V and GND come from the host connector J1,

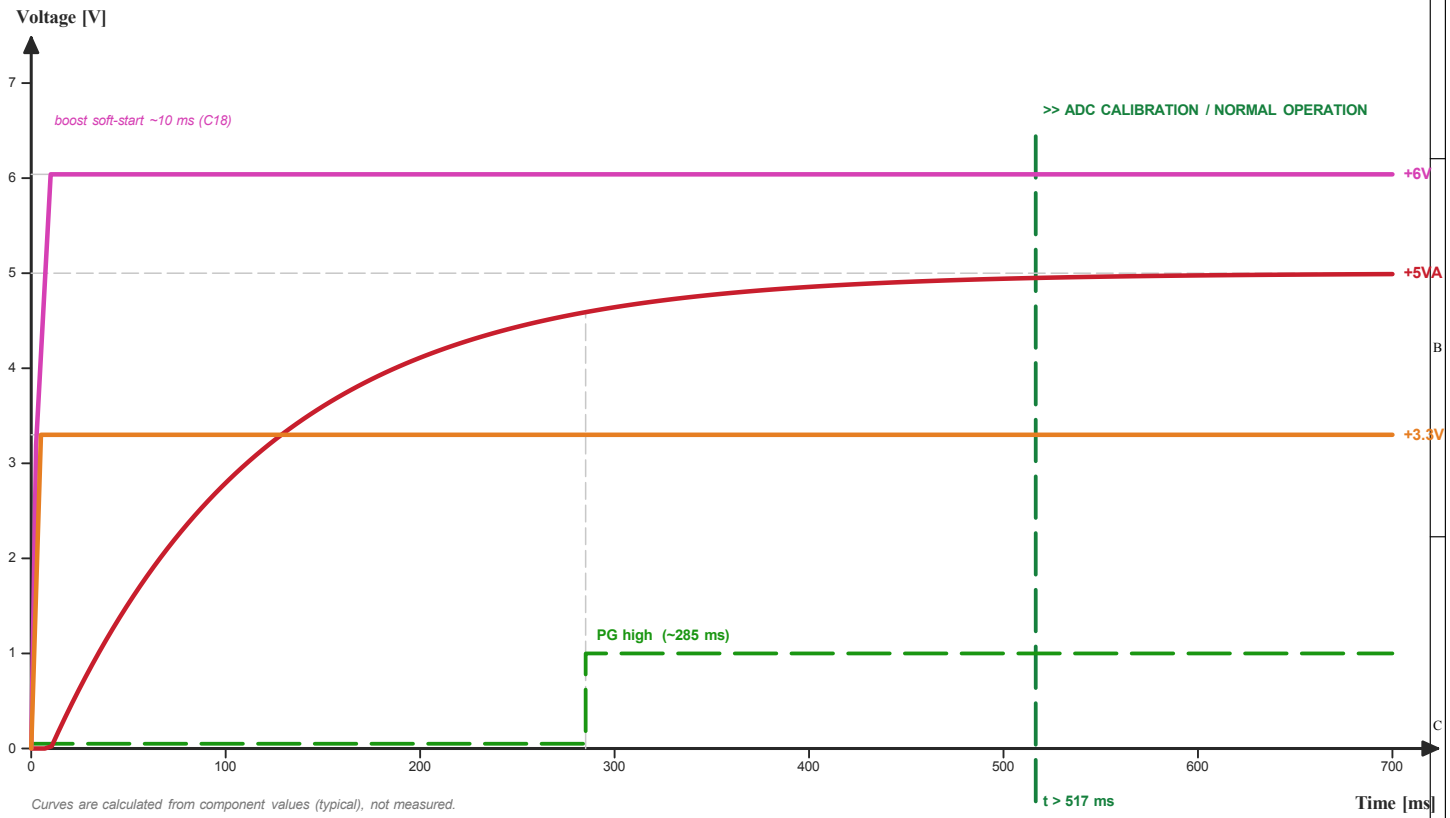
Comments: **References:** JST XH B7B-XH-AM (J1) J2/J4 21033213501, H5 WE-SHC	Company: **OPEN THRUST LAB**		Variant:	Git Hash: 349b121
	Board Name: **AD7190**		Project Name: **OPEN_WEIGHT**	
Sheet Title: Connectors	File Name: Connectors.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Connectors/		Reviewer:	Size: **A3**	Sheet: **8** of **10**

[9] Power - Sequencing

NAME	SOURCE	LEVEL
+3.3V	Host (J1)	3.3 V (host)
+6V	TPS61086 IC2	6.04 V
+5VA	LT3042 U2	5.00 V
PG	LT3042 U2	> 4.59 V

Values from the schematic:
+6V = 1.238 V x (1 + R17/R15) = 1.238 x (1 + 69.8k/18k)
+5VA = ISET x RSET = 100 uA x 50k (R9)
PG threshold = 0.3 V x (1 + R12/R13) = 4.59 V
LT3042 soft-start: tau = RSET x CSET = 50k x 2.2 uF
= 110 ms -> +5VA at 99 % after ~550 ms

DESIGN NOTE:
The LT3042 SET capacitor (C14 = 2.2 uF) gives a 110 ms time constant. Firmware must wait >= 600 ms after power-up (or until PG, TP17) before the AD7190 reset and internal calibration. To start faster, reduce C14 (e.g. 0.47 uF -> tau = 23.5 ms) at the cost of slightly more noise.



Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: 349b121
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 9 of 10

[10] Revision History

Version Unreleased

Added

- AD7190 front-end, 2 load-cell inputs
- +3.3V -> TPS61086 6 V -> LT3042 +5VA
- Self-contained libs and 3D models (lib/)
- JLCPCB 4-layer rules and stackup
- LTspice simulations (Simulation/LTspice)
- openEMS layout coupling (Simulation/openEMS)

Changed

- Renamed to OPEN_WEIGHT, KiCad 10

Fixed

- Connectors not linked to ADC in schematic
- AD7190 solder-mask bridges

	Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: 349b121
		Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 10 of 10